EE2004 Digital Signal Processing

Tutorial 7 Answer Key

1. (a) Zeros: $\pm\frac{1}{2} \pm j\frac{\sqrt{3}}{2},\ \pm 1$
   Poles: ± 1
   The poles at ± 1 get cancelled.
   (b) $H(e^{j\omega}) = 1 + e^{-j2\omega} + e^{-j4\omega}$. To get exact plot, use the MATLAB command, `freqz([1 0 1 0 1],1,1000)`

2. Proof

3. If $0 < a < 1$, then $\omega_{3\,\mathrm{dB}} = \cos^{-1}\left[\dfrac{4a - a^2 - 1}{2a}\right]$

4. Proof